

updates the DOM when data changes, making it ideal for interactive applications.

- **Component-based system:** Like React, Vue.js uses components to organise the application's user interface.
- **Lightweight:** It is smaller and easier to learn compared to more extensive frameworks like Angular or React.

Use cases: Vue.js is best suited for lightweight SPAs, prototyping, and projects where simplicity and ease of integration are key.

Pros: Vue.js is easy to learn and has a flexible, modular structure.

Cons: Its ecosystem is smaller compared to React, which can make finding support or resources more challenging.

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Lodash

Lodash is a utility library that simplifies working with arrays, objects, and strings. It offers a range of methods to perform common tasks like deep cloning and data manipulation.

Key features

- **Utility functions:** Lodash offers an array of functions that simplify common tasks like filtering arrays, cloning objects, and performing deep comparisons.
- **Performance optimised:** Its functions are optimised for performance, especially when handling large datasets.

Use cases: Lodash is useful for simplifying code logic, improving performance, and handling complex data manipulations.

Pros: It is packed with utility methods that save development time.

Cons: With the addition of modern JavaScript features, many of Lodash's functionalities are now built into JavaScript, reducing the need for this library.

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Three.js

Three.js is a JavaScript library for creating 3D graphics in the web browser using WebGL. It makes 3D rendering accessible for web developers without requiring deep knowledge of WebGL.

Key features

- **3D rendering:** Three.js simplifies the process of creating 3D environments, objects, and animations in the browser.
- **Animation support:** It includes built-in functions for handling 3D animations, shadows, and lights.
- **Robust API:** The library offers extensive documentation and APIs for advanced visual effects.

Use cases: Three.js is commonly used for interactive 3D applications, games, and augmented/virtual reality (AR/VR) projects.

Pros: It enables the creation of stunning 3D visuals that work across all modern browsers.

Cons: It can cause performance bottlenecks on low-end devices and may require optimisation for smooth experiences.

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Moment.js

Moment.js is a library for parsing, validating, manipulating, and formatting dates and times in JavaScript.

Key features

- **Locale support:** Moment.js allows easy localisation for date formats and time zones.
- **Relative time:** Developers can display time in relative formats, like "5 minutes ago."
- **Time zone handling:** It helps manage dates and times across various time zones.

Use cases: Moment.js is essential in applications where date and time manipulation are crucial, such as scheduling apps or event management systems.

Pros: It is feature-rich and offers comprehensive support for time and date formatting.

Cons: It is now considered a legacy library, with modern alternatives like Day.js offering similar functionality with a smaller footprint.

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Chart.js

Chart.js is a simple yet flexible JavaScript library for creating charts and graphs. It offers easy-to-use APIs and supports a variety of chart types.

Key features

- **Predefined chart types:** Includes bar charts, line charts, radar charts, and more.
- **Responsive design:** Chart.js automatically adjusts chart dimensions based on screen size.
- **Extensibility:** Developers can customise chart elements, labels, and tooltips.

Use cases: Chart.js is perfect for dashboards, reports, and applications requiring basic data visualisation.

Pros: It is easy to implement and use, with beautiful and responsive chart visuals.

Cons: It lacks advanced customisation options for more complex visualisations.

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Axios

Axios is a promise-based JavaScript library for making HTTP requests. It simplifies the process of interacting with REST APIs and handling asynchronous data fetching.